

Board Important – Chapter 6: Anatomy of Flowering Plants

Typical weightage & high-yield topics

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Karnataka 1st PUC Blueprint

Typical Annual Exam Weightage

Topic	Usual Question Type	Marks	Priority
T.S. of Dicot Root / Monocot Root	Diagram + description (5 M) or short note	3–5 M	Top Priority
T.S. of Dicot Stem / Monocot Stem	Diagram + description (5 M) or short note	3–5 M	Top Priority
T.S. of Dicot Leaf / Monocot Leaf	Diagram + description or difference (2–3 M)	2–3 M	Top Priority
Open vs Closed / Conjoint vs Radial VB	Short note / 2 mark definition	2 M	High Priority
Differences: Dicot vs Monocot (root/stem/leaf)	Difference table, any 4 points	4–5 M	Top Priority
Epidermal tissue system, stomata, trichomes	Very short answer / MCQ	1–2 M	High Priority

High-Yield Definitions to Master Cold

- **Casparian strips** — waxy suberin deposits on radial/tangential walls of endodermal cells.
- **Exarch / Endarch** — protoxylem towards periphery (roots) vs towards centre (stems).
- **Bulliform cells** — large empty adaxial epidermal cells in grasses that control leaf rolling.
- **Conjunctive tissue** — parenchyma lying between xylem and phloem in a vascular bundle.
- **Bundle sheath** — parenchymatous layer surrounding vascular bundles in leaves/monocot stems.

Model Answers – Set A

Frequently asked 5-mark descriptive questions

Q1. Describe the internal structure (T.S.) of a dicot root, with reference to a sunflower root. (5 M)

A dicot root T.S. shows, from outside to inside: **(i) Epiblema** — outermost single layer of barrel-shaped parenchyma cells, many producing unicellular root hairs. **(ii) Cortex** — several layers of thin-walled, round/oval parenchyma cells with intercellular spaces. **(iii) Endodermis** — innermost single barrel-shaped layer, without intercellular spaces, bearing waxy Casparian strips on radial/tangential walls. **(iv) Stele** — all tissue inner to the endodermis: a single-layered **pericycle** (initiates lateral roots & vascular cambium), a **radial** vascular bundle with 2–4 **exarch** xylem patches alternating with phloem, and a small, inconspicuous central **pith**.

Q2. Draw a labelled diagram of T.S. of a monocot stem and describe its structure. (5 M)

From outside to inside: **(i) Epidermis** — single layer with thick cuticle. **(ii) Hypodermis** — few layers of **sclerenchyma**, providing mechanical strength. **(iii) Ground tissue** — large, undifferentiated parenchymatous region (cortex + pith not distinguished) with numerous **scattered** vascular bundles, more crowded towards the periphery. Each bundle is **conjoint and closed** (no cambium), surrounded by a sclerenchymatous **bundle sheath**, with **endarch** xylem in a Y/V shape, phloem below it, and prominent water-containing cavities where protoxylem lies (phloem parenchyma is absent). Pith is not demarcated.

Model Answers – Set B

Frequently asked difference / short-note questions

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Q3. Differentiate between dicot leaf and monocot leaf. (Any four points, 4 M)

Feature	Dicot Leaf (Dorsiventral)	Monocot Leaf (Isobilateral)
Mesophyll	Differentiated into palisade & spongy parenchyma	Not differentiated
Stomata	More on abaxial (lower) surface	Equal on both surfaces
Venation	Reticulate	Parallel
Special cells	Absent	Bulliform cells present on adaxial epidermis

Q4. What is a vascular bundle? Explain its open and closed types with examples. (3 M)

A vascular bundle is a strand of conducting tissue made of **xylem and phloem**. It is called **open** when a strip of cambium lies between the xylem and phloem — this cambium can later divide to produce secondary xylem/phloem, giving **secondary growth**; open vascular bundles are found in **dicotyledonous** stems. It is called **closed** when no cambium is present between xylem and phloem, so no secondary growth occurs; closed vascular bundles are found in **monocotyledonous** stems.

Q5. Mention any four differences between dicot stem and monocot stem. (4 M)

Feature	Dicot Stem	Monocot Stem
Hypodermis	Collenchymatous	Sclerenchymatous
Vascular bundles	Arranged in a ring, limited in number	Scattered, numerous
Cambium	Present (open)	Absent (closed)
Pith	Present, well developed	Absent / not demarcated